

ENDERBY ISLAND, New ZealandWMO: **939290**Lat: **50.493S**Lon: **166.296E**Elev: **131**StdP: **14.63**Time Zone: **12.00 (NZL)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.9	34.8	24.8	19.0	35.6	26.6	20.7	36.8	44.6	47.6	41.0	46.5	17.8	210	0.983

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.4	56.1	53.5	55.1	52.8	54.4	52.2	54.8	55.3	53.9	54.5	53.1	53.8	20.1	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.5	63.4	54.9	53.5	61.2	54.1	52.7	59.3	53.4	23.2	55.7	22.6	54.5	22.1	53.9	59.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 41.2	37.1	33.8	29.5	57.8	1.9	1.2	28.1	58.6	27.0	59.3	26.0	59.9	24.6	60.8		
(5)			27.8	56.0	1.9	1.5	26.4	57.1	25.3	57.9	24.3	58.7	22.9	59.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.3	50.2	50.3	49.3	47.5	45.5	43.3	43.0	43.5	44.1	44.8	45.9	48.7		
	DBStd	3.98	2.25	2.36	2.56	2.86	3.60	3.67	3.57	3.13	3.19	2.74	2.89	2.61		
	HDD50	1455	25	23	44	82	143	200	217	202	178	163	123	55		
	HDD65	6820	460	412	488	524	606	650	682	667	628	627	572	504		
	CDD50	111	31	31	21	8	3	0	0	0	0	0	1	16		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	18.7	18.3	18.3	18.5	19.2	19.1	18.8	18.7	18.6	19.1	19.6	18.9	17.2		
	PrecAvg	39.7	3.9	3.1	4.0	3.3	3.7	2.9	3.1	2.7	3.0	3.2	3.0	3.2		
(15) Precipitation	PrecMax	47.4	8.0	6.7	6.4	6.9	5.6	7.1	5.2	5.0	6.0	6.7	6.7	6.1		
	PrecMin	31.8	1.6	0.7	2.1	0.7	0.9	0.0	0.5	0.0	0.5	0.6	0.2	1.0		
	PrecStd	4.8	1.7	1.5	1.1	1.4	1.1	1.5	1.1	1.3	1.4	1.3	1.5	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	57.4	56.4	54.6	53.1	51.0	50.3	50.3	51.0	52.4	53.7	56.7		
		MCWB	54.0	55.0	55.0	53.2	51.8	49.5	48.9	48.5	49.2	49.8	50.5	53.2		
	2%	DB	55.9	56.0	55.1	53.6	52.1	50.1	49.3	49.4	50.1	51.1	52.2	54.9		
		MCWB	52.7	53.7	53.7	52.3	51.0	48.9	48.1	47.8	48.5	48.9	49.6	52.0		
	5%	DB	54.8	54.9	54.2	52.8	51.3	49.4	48.6	48.6	49.3	50.1	51.2	53.9		
		MCWB	52.0	52.8	52.7	51.5	50.3	48.2	47.5	47.2	47.8	48.0	48.9	51.3		
	10%	DB	53.9	54.1	53.3	52.0	50.5	48.5	47.9	47.9	48.5	49.2	50.3	52.9		
		MCWB	51.5	52.1	51.8	50.7	49.4	47.3	46.8	46.5	47.0	47.2	48.3	50.6		
	0.4%	WB	55.7	56.2	55.7	53.9	52.4	50.3	49.5	49.3	50.2	51.0	51.9	54.6		
		MCDB	56.4	56.7	55.9	54.1	52.6	50.7	49.8	49.7	50.3	51.5	52.6	55.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	54.0	54.8	54.4	52.9	51.4	49.4	48.6	48.4	49.1	49.7	50.6	53.3		
		MCDB	54.8	55.4	54.8	53.3	51.8	49.8	49.0	48.9	49.5	50.4	51.4	54.1		
	5%	WB	53.1	53.9	53.3	52.0	50.6	48.5	47.8	47.6	48.3	48.7	49.9	52.3		
		MCDB	54.0	54.5	53.9	52.5	51.1	49.1	48.4	48.3	48.9	49.5	50.7	53.2		
	10%	WB	52.1	52.9	52.3	51.1	49.7	47.5	47.0	46.8	47.4	47.8	49.0	51.3		
		MCDB	53.3	53.7	53.0	51.8	50.4	48.4	47.7	47.7	48.2	48.8	49.9	52.4		
	5% DB	MDBR	6.4	6.4	5.9	5.7	5.3	5.5	5.2	5.4	5.9	6.9	6.8	6.6		
		MCDBR	7.1	7.0	6.1	5.7	5.4	5.4	4.9	5.2	5.8	7.0	7.0	7.2		
		MCWBR	5.9	6.2	6.1	6.3	6.0	6.1	5.5	5.5	5.6	6.6	6.3	6.0		
		MCWBR	6.3	6.1	5.5	5.6	5.3	5.4	4.6	5.1	5.2	6.3	6.1	6.1		
(35) Mean Daily Temperature Range	5% WB	MCWBR	6.0	6.3	6.1	6.5	6.1	6.2	5.3	5.5	5.5	6.5	6.2	5.6		
	taub		0.346	0.334	0.330	0.319	0.316	0.308	0.311	0.322	0.346	0.336	0.336	0.337		
		taud	2.509	2.560	2.571	2.563	2.464	2.386	2.445	2.505	2.448	2.507	2.518	2.537		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		305	299	282	254	217	197	215	248	270	295	307	311		
		Edh at Noon	34	30	26	21	18	16	18	22	29	31	33	33		
	RadAvg		1704	1361	930	553	293	198	249	467	803	1287	1568	1783		
(45) All-Sky Solar Radiation	RadStd		101	82	58	30	21	10	19	35	47	51	104	113		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.65	N/A	+1.19	+0.49	+0.85	+1.03	N/A	-176	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page